Neurocognitive Linguistics: Foundations and Implications for CCF Version 7

Purpose

This document summarizes the Neurocognitive Linguistics course presentations by Sydney M. Lamb in:

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/mnt/ssd/ely/linguistics/cursos/NeurocognitiveLinguistics/Curso/
```

The folder contains 21 slide decks covering neurolinguistic evidence, cortical columns, relational networks, functional webs, cardinal nodes, cognitive maps, perception, linguistic structure, and learning. Three decks are also available as PDFs. The summary focuses on the ideas that matter for the next CCF discussion: how to abstract a cortical column as the framework's basic unit while preserving Lamb's broader account of language and cognition.

The document also incorporates three recent language-neuroscience papers and two cortical-column study artifacts: a circuit survey and an interactive population-level visualization. These additions update classical language localization, evaluate evidence relevant to cardinal nodes, and relate laminar column circuitry to NCL before CCF chooses an implementation simplification.

The source materials mix several evidential statuses. This document distinguishes:

- **observations and established constraints** used by Lamb;
- **Neurocognitive Linguistics/RNT hypotheses** proposed from those constraints;
- **possible CCF consequences**, which remain design proposals;
- **scientific cautions**, where a slide's claim is stronger than current evidence warrants.

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1. The central project

Neurocognitive Linguistics studies the **physical linguistic system of an individual**, not primarily sentences, grammars, or texts considered as external symbolic products.

Its starting chain of reasoning is:

```
people can produce and understand language
→ an operational linguistic system must exist
→ that system is physically realized in the nervous system
→ the cerebral cortex is a dynamic network
→ linguistic structure must be realizable as network structure
```

The resulting theory is constrained by three forms of plausibility:

1. **Operational plausibility** — the proposed system must actually produce, recognize, and learn linguistic behavior.
2. **Developmental plausibility** — it must be possible for a child to develop the system from experience.
3. **Neurological plausibility** — its units, connections, and operations must be compatible with known nervous-system organization.

This reverses the usual explanatory direction. A symbolic description of a sentence may be linguistically useful, but it is not yet an account of the system that produced or understood it. Neurocognitive Linguistics asks what network must exist for that behavior to be possible.

Core thesis

> Linguistic and conceptual structure consists of relations among network locations. The locations do not contain symbols; their functions are determined by their connectivity, thresholds, activation histories, and positions in larger networks.

Language is not isolated from cognition. Phonological, lexical, grammatical, conceptual, perceptual, and motor structures belong to one larger human information network.

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2. Sources of evidence

The course develops its position by combining several evidence streams.

2.1 Linguistic behavior

Observable texts, comprehension, production, slips, lexical learning, polysemy, morphological analysis, and grammatical constructions constrain the hidden system. Lamb follows Hjelmslev's idea of **catalysis**: infer an unobserved relational organization capable of producing the observed behavior.

2.2 Aphasiology and lesion evidence

Broca, Wernicke, Lichtheim, Geschwind, Luria, Damasio, Goodglass, and others provide evidence that:

- complex language behavior depends on cooperation among several areas;
- damage can selectively disrupt phonological recognition, production, repetition, naming, writing, reading, semantic categories, or relational markers;
- deficits vary continuously and rarely fit one exact syndrome;
- speaking and "understanding" are complex functions, not elementary functions located in one center;
- written form can connect to meaning independently of phonological form;
- production normally uses perceptual/recognition systems for planning and monitoring.

The presentations endorse a qualified Wernickean principle:

> Local areas perform relatively limited functions; complex behavior results from interconnected areas operating together.

2.3 Neuroanatomy

The major constraints emphasized are:

- cortex is composed of interconnected excitatory and inhibitory neurons;
- pyramidal neurons provide most long-range cortical output;
- local and distant connectivity differ in density and function;
- corticocortical pathways are commonly reciprocal;
- cortex is organized in layers and columnar populations;
- primary areas, long-range tracts, and broad hemispheric organization provide inherited constraints;
- experience refines organization within those constraints.

2.4 Perceptual neuroscience

Mountcastle's work on somatosensory, visual, and auditory organization is used as the principal bridge from cortical physiology to language. The course extrapolates local specificity, adjacency, competition, and columnar organization from perceptual cortex to associative linguistic and conceptual cortex.

2.5 Functional mapping

The decks review electrical stimulation, lesion mapping, Wada testing, EEG/ERP, PET, fMRI, MEG/MSI, TMS, and diffusion imaging. Lamb repeatedly warns that an area that "lights up" is not automatically the exclusive store or center of a function. Methods differ in spatial resolution, temporal resolution, sensitivity, and inference problems.

MEG is especially valued for following activation over time, but source localization remains an inverse problem and detects synchronized populations rather than individual columns.

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3. From neurons to cortical columns

3.1 The proposed operational node

Lamb adopts Mountcastle's statement that the effective cortical unit is not an isolated neuron but a group of cells with similar response properties and connectivity. In the course's abstraction:

```
small scale: a cortical column is a network of neurons
large scale: that column behaves as one node in a cortical network
```

The presentations distinguish:

Unit	Course description
Minicolumn	Narrow, vertically organized population spanning cortical layers; approximately 70–110 neurons in the cited account.
Functional column	Experience-dependent contiguous group of minicolumns performing a shared function; variable in size.
Maxicolumn	Larger bundle of roughly a hundred minicolumns.
Hypercolumn	Still larger organization containing related maxicolumns.
Supercolumn	Generic term introduced in the learning deck for a columnar population of any relevant scale.

For Lamb, many nodes in a linguistic relational network probably correspond to **functional columns or larger supercolumns**, not necessarily one anatomical minicolumn.

3.2 Minimal columnar operations

The local column does not store a word, feature, proposition, or executable rule. It performs a small set of operations:

1. **Integration** — combine incoming excitation and inhibition.
2. **Thresholding** — become active to a graded degree when input is sufficient.
3. **Persistence** — recurrent local activation can remain active temporarily.
4. **Broadcasting** — send activation to many connected destinations.
5. **Competition** — local inhibitory output suppresses nearby alternatives.
6. **Plasticity** — successful co-activation changes connection strengths and thresholds.

A compact functional description is:

```
incoming activation
+ local recurrent state
− inhibition
→ graded threshold response
→ excitatory/inhibitory broadcast
→ possible learning
```

3.3 Nodal specificity

The course proposes that every functional node has a specific local function. A column can therefore serve as the node for a phonological pattern, property, conjunction, category, lexical form, or other relational value.

Specificity does **not** mean that the column independently contains everything known about that value. It means that its place in the network is dedicated to one convergence function.

3.4 Connectivity determines function

The same broad columnar machinery is proposed across associative cortex. Different regions process different information mainly because they connect to different inputs, outputs, neighbors, and long-range systems.

Lamb summarizes this as:

> What matters is not what is stored in a node, but where it is in the network and what it connects to.

The course acknowledges structural exceptions in heterotypical primary and specialized sensory areas. Its strongest uniformity claim concerns homotypical association cortex.

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4. Relational Network Theory

4.1 Relations rather than symbolic objects

In a relational-network diagram, labels such as `BOY`, `boy`, `/b/`, or `NOUN` help a human read the figure, but the labels are not components of the represented cognitive system. Removing a label leaves the network organization unchanged.

A morpheme is identified through connections to:

- phonological recognition and production;
- grammatical categories and constructions;
- conceptual and perceptual networks;
- other lexical structures.

Another node with exactly the same connections would be a redundant realization of the same function.

4.2 Relations all the way down

The system does not consist of symbolic objects joined by relations. It consists of relations to relations across levels, ending at physical interfaces:

```
higher conceptual and grammatical organization
↕
lexical and phonological relations
↕
auditory/articulatory feature relations
↕
cochlear input or muscle-control output
```

At the interface, activation crosses from the relational cortical system to receptors, muscles, and other physical systems.

4.3 Abstract and narrow notation

Lamb uses two levels of description:

- **Abstract notation** is compact, often bidirectional, and marks functional AND, OR, sequence, and precedence relationships.
- **Narrow notation** decomposes an abstract relation into directed nodes and connections closer to neural implementation.

A bidirectional abstract line generally corresponds to two distinct directed pathways. An abstract AND or OR may require several narrow nodes with different thresholds and inhibition.

This is an explicit warning against collapsing functional description directly into neurons. Multiple levels are valid if their translations are stated.

4.4 Core relational operations

The presentations derive the following network operations:

- **OR:** one of several alternatives can activate the next node;
- **AND:** several inputs must jointly satisfy a higher threshold;
- **ordered AND:** components must occur in a learned sequence;
- **precedence/ordered OR:** one marked alternative blocks or overrides a default;
- **WAIT:** recurrent activation keeps an earlier component active until a later component arrives;
- **competition:** mutually inhibitory alternatives sharpen distinctions;
- **graded satisfaction:** thresholds and connection strengths support partial/prototypical activation.

These are network configurations, not symbolic instructions interpreted by a separate processor.

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5. Functional webs

5.1 Definition

A **functional web** is a distributed subnetwork whose components jointly realize a concept, word, percept, action, or other learned function. A web:

- spans several cortical regions;
- contains functionally distinct subwebs;
- is hierarchically organized;
- can be activated from several entry points;
- overlaps other webs by sharing reusable nodes;
- supports pattern completion through reciprocal activation.

A CAT web can include visual, auditory, tactile, motor, affective, autobiographical, conceptual, phonological-recognition, and phonological-production subwebs.

5.2 Distinct subweb functions

Lamb modifies Pulvermüller's functional-web account. Components of a web need not have the same local response function. They can all become active when the web ignites because they are strongly interconnected, while each still performs a distinct job.

For example:

- a visual subweb responds to cat-like visual organization;
- an auditory subweb responds to meowing;
- a tactile subweb contributes fur texture;
- a conceptual subweb integrates category-level information;
- a phonological subweb recognizes or produces the relevant word form.

Their joint activity is similar at the whole-web level, but their local receptive functions differ.

5.3 Hierarchy

Each web and subweb is proposed to have a hierarchy:

```
low-level sensory or motor features
→ intermediate conjunctions and categories
→ subweb cardinal node
→ whole-web cardinal node
```

Higher nodes appear "smart" because they can recruit the processing capacity of the network beneath them. The intelligence belongs to the hierarchy, not to an intrinsically complex symbol inside the top node.

5.4 Overlap and reuse

Functional webs are not disjoint objects. The node for BLACK can participate in the webs for charcoal, cats, clothing, night, ink, and many other concepts. The same phonological component can participate in many words.

This avoids copying the same property or component into every representation. It also means that a "web" is a contextually selected view of one global network, not necessarily an encapsulated module.

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6. Cardinal nodes

6.1 Definition

A **cardinal node** is the highest convergence node of a functional web or subweb. It is unique to the represented function but depends on its supporting distributed hierarchy.

The result is simultaneously:

- **local representation:** one addressable convergence node;
- **distributed representation:** a large supporting network containing the content.

This is Lamb's "sophisticated grandmother node," contrasted with a naive grandmother cell that recognizes an individual by itself.

6.2 Why Lamb considers cardinal nodes necessary

The presentations offer four arguments:

1. **Hierarchy and specificity:** a finite hierarchy has a highest convergence level, and that node has a specific function.
2. **Ignition:** activation from one subweb must be able to recruit the rest of the distributed web.
3. **Learning:** repeated co-activation naturally recruits a higher node representing the conjunction.
4. **Arbitrary linguistic signs:** one phonological form must connect coherently to one conceptual organization without arbitrary links between unrelated fragments of form and meaning.

The lexical link is therefore approximately:

```
[phonological-form cardinal] ⇄ [conceptual cardinal]
```

rather than many arbitrary connections such as one phoneme linking to shape and another phoneme linking to motor use.

6.3 Generic and individual concepts

The cognitive-map deck presents a hierarchy such as:

```
ANIMAL → MAMMAL → CARNIVORE → CANINE → DOG
       → TERRIER → JACK RUSSELL TERRIER → EDDIE
```

Each more specific category reuses the broader category's network and adds distinguishing connections. This directly supports the proposed CCF treatment:

```
[TOM cardinal] → [CAT cardinal] → broader category and feature networks
```

TOM need not reconnect independently to every generic feline property. Its web contains its link to CAT plus individual-specific appearance, behavior, relations, and episodes.

The category connection should be asymmetric in operation: TOM strongly recruits CAT; generic CAT activation should not indiscriminately ignite every cat individual.

6.4 Damage and graceful degradation

The slides raise but do not fully answer what happens if a cardinal node is damaged while lower web structure survives. The theory predicts impaired coordination, naming, cross-modal ignition, or category access, while some feature-level knowledge may remain reachable through alternative routes. This is a testable prediction, not established proof of cardinal nodes.

6.5 Recent evidence relevant to cardinal nodes

Recent intracranial work makes the local-plus-distributed proposal more plausible without demonstrating Lamb's exact cardinal-node architecture.

Lakretz et al. (2024) recorded more than 1,000 single neurons in 21 neurosurgical patients. They found neurons with sharply selective responses to orthographic features, letters at approximate positions, phonemes, and phonological features. They also found modality-independent neurons in middle temporal and inferior frontal areas that responded to both heard and written linguistic input. This establishes that highly specific local responses can coexist with distributed population processing. It supports **nodal specificity as a possible neural organization principle**, but a selective neuron is not necessarily a cortical column, a hierarchy apex, or a cardinal for a whole concept.

The same paper discusses earlier evidence for sparse, modality-invariant "concept cells" in the human medial temporal lobe. Such cells can respond to a particular person or concept across pictures and names, can reactivate during recall, and can respond when a pronoun refers to their preferred concept. These findings are compatible with a local access point into a distributed representation. However:

- the recorded cells are mainly in hippocampal and medial-temporal episodic-memory systems, not identified neocortical conceptual cardinals;
- sparse coding still involves populations and does not imply one uniquely necessary cell;
- responsiveness does not by itself establish ignition of an entire functional web;
- neither the existence of a hierarchy apex nor the claim of exactly one cardinal per concept was tested.

The updated position for CCF is therefore:

> A cardinal is a plausible computational role—a compact, addressable convergence and reactivation point—but current evidence does not require it to be one neuron, one anatomical minicolumn, or one irreplaceable node.

A CCF cardinal may initially be realized as one simulated column, but the architecture should permit a small redundant cardinal population, several context-sensitive access points, and graceful rerouting. The scientific claim to test is whether such convergence points improve binding, cross-modal access, pattern completion, lexical grounding, and damage tolerance—not whether a literal CAT column has already been observed.

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7. Cognitive maps and categories

7.1 Cognitive-map hypothesis in the course

Lamb uses **cognitive map** in a broad topographic sense: cortical knowledge is organized so that nearby locations have related functions. This is established for somatotopic and tonotopic maps and hypothesized for phonological and conceptual systems.

In this summary, a cognitive map means cortical topography and learned functional proximity. It does not imply a separate latent transition processor.

7.2 Proximity principle

The course proposes:

> More closely related functions tend to occupy more closely related cortical locations.

Three factors shape location:

1. **economic constraints:** not every column can connect directly to every other column;
2. **genetic organization:** primary areas and major long-range tracts constrain broad placement;
3. **experience:** learning recruits available nearby structure and refines individual organization.

An integrating function tends to develop near the systems whose outputs it combines. Competitors tend to be neighbors because they share broad function but differ in a narrower dimension.

7.3 Categories as graded network organizations

The presentations reject small necessary-and-sufficient feature lists for most natural concepts. Categories exhibit:

- fuzzy boundaries;
- graded membership;
- prototypes and peripheral cases;
- language- and culture-dependent organization;
- hierarchical subcategories;
- effects on perception and thought.

A category cardinal receives weighted activation from many properties. Different subsets can satisfy its threshold to different degrees:

```
strong, frequent property pattern → rapid, strong cardinal activation
partial or atypical pattern       → weaker, slower activation
insufficient pattern              → no stable category activation
```

Prototypicality follows from weighted connections, activation strength, frequency, and threshold satisfaction rather than from a symbolic prototype record.

7.4 Columnar subdivision and category learning

The course hypothesizes that broad distinctions recruit a large supercolumn first. Further learning subdivides it into smaller competing functional columns:

```
broad category
→ categories
→ subcategories
```

```
→ finer distinctions
```

This is presented as an extrapolation from perceptual column organization, not direct evidence that conceptual taxonomies literally map one-to-one onto anatomical hypercolumns.

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8. Perception, imagery, and inference

8.1 Macroperception

Perception is not treated as a purely bottom-up, unimodal posterior process. Lamb distinguishes:

- **microperception/recognition:** local bottom-up integration within one modality;
- **macroperception:** multimodal, bidirectional, conceptually informed perception involving sensory, conceptual, and motor networks.

Examples include the McGurk effect, motor activation during hand recognition, mirror-neuron phenomena, and language effects on category discrimination.

8.2 Bidirectional processing

Reciprocal cortical connections allow partial sensory evidence to activate a cardinal and then reactivate expected properties:

```
partial features → category cardinal → associated properties
```

This supplies a network account of:

- imagery;
- pattern completion;
- inference beyond current input;
- perceptual expectations;
- production monitoring;
- context-sensitive interpretation.

The theory does not require little pictures, sounds, or words inside the brain. Imagery reactivates some of the same structures used during perception and action.

8.3 Reverberation

Activation circulating through an established web strengthens and stabilizes the whole pattern. Real words can produce stronger late activity than phonologically legal nonwords because only real words recruit entrenched semantic and lexical webs.

Reverberation is also proposed for short-term maintenance across recognition and production systems.

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9. Linguistic organization

9.1 Several interacting strata

The course distinguishes but connects:

- auditory and articulatory features;
- phonological recognition;
- phonological production;
- somatosensory monitoring;
- graphic recognition and production;
- lexical/morphological networks;
- grammatical constructions;
- conceptual, perceptual, and motor meaning.

These are not a rigid processing pipeline. Activation can travel in either direction and several levels can operate concurrently.

9.2 Phonology

Phonological units are network locations defined through lower auditory features and higher lexical combinations. The presentations emphasize transitions and demisyllables rather than assuming each alphabetic-looking phoneme is a perceptual object.

Shared components are represented once and connected into many larger structures. Competition among neighboring alternatives sharpens categorical perception.

9.3 Lexicon and morphology

A learned word or phrase has a cardinal only when experience has entrenched it as a unit. Complex forms can support parallel analyses with different connection strengths. Examples include `hamburger`, `hotdog`, and morphologically structured forms such as `merciless`.

The network naturally represents:

- degrees of entrenchment;
- productive and lexicalized analyses in parallel;
- "shadow meanings" from partially active constituents;
- recurring semantic components such as DIE in *die*, *kill*, *murder*, and *assassinate*;
- polysemy through one lexical cardinal broadcasting toward several meaning networks;
- puns when context supports more than one meaning.

9.4 Grammar

Syntax is represented by relational constructions, combinations, alternatives, and sequencing—not stored sentence templates or rules interpreted by another mechanism.

An ordered conjunction can recognize a subject-like constituent followed by a predicate-like constituent. Linked constructions form larger grammatical networks. Grammatical processing is treated as higher-level recognition using the same broad operations as perception.

Aphasic loss of relational markers and grammatical production is used as evidence that syntax depends on distributed phonological, sequencing, memory, and frontal systems rather than one indivisible grammar center.

9.5 Meaning and language

Conceptual content is distributed over sensory, motor, social, affective, and higher associative systems. Lexical structures provide learned routes into that content. Language also reshapes cognition because frequently used lexical and grammatical categories repeatedly strengthen selected pathways.

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10. Learning

10.1 Basic rule

The learning decks use a refined Hebbian principle:

> A connection strengthens when its activation contributes, together with other active inputs, to successful activation of a target node.

Independent firing can weaken mutual connections. Learning can involve:

- biochemical synaptic change;
- dendritic-spine growth;
- formation or removal of synapses;
- threshold adjustment;
- recruitment and subdivision of columnar populations.

10.2 Abundance hypothesis

The system begins with abundant weak or silent possibilities:

- latent nodes/columns;
- latent excitatory and inhibitory connections;
- many more possible pathways than will become dedicated.

Learning is selective rather than constructing arbitrary wiring from nothing. Repeated co-activation recruits a latent higher node for a conjunction, strengthens successful paths, and raises or calibrates its threshold.

10.3 Bottom-up hierarchy formation

If A and B repeatedly co-activate and jointly recruit C, C becomes a node for the conjunction AB. C can then participate in still higher learning. The resulting organization is hierarchical and developmental:

```
features → conjunctions → categories → complex webs
```

10.4 Subdivision and differentiation

Early learning may recruit a large supercolumn for a broad distinction. Later experience subdivides it into more specific functional columns, while lateral inhibition sharpens alternatives.

The presentations explicitly leave the learning of appropriate lateral inhibition incompletely explained.

10.5 Entrenchment

New learning and continued practice belong to the same continuum. Repeated use strengthens connections, lowers effective access cost, and can stabilize an analysis as a unit. Every use slightly changes the system.

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11. Hemispheres and distributed language

The course rejects a simple "language is only in the left hemisphere" account.

It emphasizes:

- left-hemisphere advantages for sharply differentiated phonological and grammatical processing;
- bilateral participation in speech perception and conceptual processing;
- right-hemisphere contributions to broad semantic activation, connotation, discourse inference, metaphor, prosody, and music;
- reciprocal interhemispheric connectivity;
- substantial individual variation and plasticity.

The proposed explanation is again connectivity, granularity, inhibition, and network organization rather than two wholly different kinds of processor.

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12. Recent evidence: updating the language architecture

12.1 Papers and evidential status

Three recent papers in the accompanying library refine the course account:

1. **Fedorenko, Ivanova, and Regev (2024), "The language network as a natural kind within the broader landscape of the human brain," *Nature Reviews Neuroscience*, 25, 289–312, doi:10.1038/s41583-024-00802-4.** This is a peer-reviewed synthesis of functional localization, lesion, stimulation, connectivity, and intracranial evidence.
2. **Lakretz et al. (2024), "Modality-Specific and Amodal Language Processing by Single Neurons," bioRxiv, doi:10.1101/2024.11.16.623907.** This is a large intracranial study, but the supplied version is a preprint explicitly marked as not peer reviewed. Its single-neuron findings are important but should remain provisional pending review and replication.
3. **Casto, Ivanova, Fedorenko, and Kanwisher (2025), "What does it mean to understand language?", arXiv:2511.19757v1.** This is an opinion/research-program preprint. It synthesizes existing results and proposes testable "exportation" hypotheses; it does not report a definitive demonstration of all proposed pathways.

The papers do not invalidate NCL's general premises that language is physical, relational, distributed, learned, bidirectional, and embedded in a larger cognitive network. They replace coarse named centers with a more precise set of interacting functional networks and sharpen the boundary between linguistic form processing and grounded conceptual understanding.

12.2 What replaces the classical Broca–Wernicke model

The modern picture is neither "one center per faculty" nor "the whole brain does language without specialization." It is a **nearly decomposable system of specialized but interacting networks**.

Core high-level language network

Fedorenko et al. identify a strongly interconnected, mostly left-lateralized network of lateral temporal and inferior/middle frontal functional areas. Its principal properties are:

- it participates in both comprehension and production;
- it is largely independent of input/output modality, responding to spoken, written, and signed language;
- it is stable within an individual but varies substantially in exact anatomical position across individuals;
- it is causally important for language and selective for language over arithmetic, music, general executive tasks, nonverbal event understanding, and theory-of-mind reasoning;
- it processes regularities at several scales, including sublexical patterns, morphology, lexical meaning, and phrase/sentence composition;
- lexical-semantic and syntactic/compositional responses are strongly integrated throughout the network rather than assigned to cleanly separate syntax and semantics centers;
- its different macroscopic regions have surprisingly similar response profiles, although intermingled neural populations within them can have distinct temporal windows and computations.

This is strong modern support for Lamb's Wernickean premise: complex linguistic behavior comes from an interconnected system, and a gyrus-sized area should not be equated with a complete cognitive function. It also supports NCL's use of shared relational operations across word and construction sizes.

Specialized perceptual interfaces

The core network is functionally distinct from input-specific systems:

- bilateral superior temporal speech-perception areas analyze speech-specific spectrotemporal and phonological structure before lexical meaning;
- the visual word-form area (VWFA) in ventral occipitotemporal cortex analyzes familiar written forms before they are linked to meanings;
- general auditory and visual cortices supply still earlier sensory processing.

Lakretz et al. provide convergent single-neuron evidence for this separation. Fusiform neurons responded selectively to written input, including fork-like letter features and particular letters in approximate word positions. Heschl's-gyrus and superior-temporal neurons responded selectively to heard speech, including particular consonants and diphthongs. In contrast, some middle-temporal and inferior-frontal neurons responded to both modalities.

The findings refine, rather than reject, NCL strata:

```
sensory signal
→ modality-specific speech or orthographic subweb
→ amodal high-level linguistic network
↔ extra-linguistic conceptual and action systems
```

This is not a rigid one-way pipeline. Both reviews preserve top-down effects, recurrence, and bidirectional information flow. However, CCF should not merge early phonological/orthographic columns with lexical or conceptual cardinals.

Specialized output interfaces

The label **Broca's area** has referred to several incompatible anatomical and functional objects. In the Fedorenko et al. synthesis, a functionally identified posterior inferior-frontal area participates in a speech-articulation network and prepares articulatory motor plans. Ventral sensorimotor cortex executes those plans and represents articulator states. Nearby frontal regions belong to the modality-independent high-level language network. Macroanatomical averaging can blur these distinct populations.

The label **Wernicke's area** is similarly ambiguous. A bilateral superior temporal speech-perception system may be close to Wernicke's original "sound-image" function, whereas high-level lexical and combinatorial language responses extend through other temporal and frontal areas. "Wernicke area = comprehension center" is therefore not an adequate implementation rule.

For CCF, `BROCA` and `WERNICKE` should never be processor types. The functional distinctions to model are:

- speech-form perception;
- orthographic-form perception;
- modality-independent lexical/construction processing;
- articulatory planning;
- sensorimotor execution and monitoring;
- their learned reciprocal routes.

Functional localization rather than anatomical labels

Association cortex is functionally heterogeneous, and functional topography varies between people. Group averaging in common anatomical coordinates can combine different neighboring systems and create misleading dissociations. Recent research therefore identifies functional areas within each participant using validated localizers and then tests those areas independently.

The CCF analogue is important: a unit's identity should be determined by its learned response profile and connectivity, not by assigning a cognitive label from a fixed coordinate or normalized name. Stable simulated IDs remain useful, but their functional interpretation should come from reproducible activation, lesion, connectivity, and transfer tests.

12.3 Local specificity and distributed processing coexist

Lakretz et al. found a progression from sharply modality-specific responses in fusiform and superior temporal cortex to modality-independent responses in middle temporal and inferior frontal regions. At the higher level, they did not find neurons selective to one isolated syntactic value; high-level features appeared more complex, intermingled, or distributed.

They also found that single-cell firing correlated more strongly with high-gamma activity measured on the same microwire than with macro contacts a few millimetres away. This suggests locally redundant coding at one scale and rapid loss of that exact redundancy over larger distances. It is compatible with local functional populations embedded in broader distributed networks, but it does not establish the boundaries of a minicolumn or functional column.

For CCF this supports a multiscale rule:

```
selective local unit or population
≠ self-contained representation

selective local unit or population
+ distributed connectivity
= possible access, feature, conjunction, or control function
```

Low-level forms may use sparse, interpretable feature selectivity. Higher-level language and cognition may rely more heavily on mixed and distributed population codes. CCF should not require every simulated column to have a human-readable one-feature label.

12.4 Language is not identical to conceptual cognition

Fedorenko et al. distinguish the core language network from systems for world knowledge and reasoning. Severe aphasia can coexist with preserved nonlinguistic event understanding, arithmetic or logic in some patients, social reasoning, and other conceptual abilities. Conversely, difficult reasoning tasks recruit multiple-demand systems that are not part of the language network.

The core language network appears to access words and constructions and combine them into a modality-independent, compositionally structured approximation of meaning. It does not by itself supply all grounded knowledge, physical simulation, social inference, episodic memory, or extended discourse organization.

Casto et al. call this **shallow understanding**, while calling the construction of grounded, context-sensitive situation models **deep understanding**. Their proposed “exportation” account states that high-level linguistic representations are passed to other systems, potentially including:

- theory-of-mind networks for beliefs, intentions, sarcasm, and pragmatics;
- intuitive-physics systems for physical prediction;
- scene and navigation systems for spatial situations;
- perceptual, motor, and affective systems;
- semantic and episodic memory systems;
- longer-timescale default-mode systems for narratives and situation models;
- multiple-demand systems when a task requires control, calculation, or explicit problem solving.

The term “exportation” should not be read as proof of a single serial message bus. Selective routing versus broad broadcasting remains an explicit open question, as do representational translation, timing, and the role of hippocampal coordination. The flow must also be bidirectional: intended meanings drive production, while world knowledge, visual context, gesture, discourse, and interlocutor knowledge influence comprehension.

This update strongly supports CCF’s separation of:

```
spoken/written form
→ lexical and constructional organization
→ grounded conceptual, perceptual, action, episodic, and structural maps
```

It also qualifies a simple NCL edge from one phonological cardinal directly to one complete conceptual cardinal. The arbitrary linguistic sign still needs stable form-to-meaning alignment, but “meaning” may involve a language-internal lexical/construction representation that provides access to several distributed, non-language conceptual systems. A lexical cardinal must not substitute for grounding.

12.5 Discourse and situation models extend beyond the core language network

The high-level language network is sensitive to words and sentence-level composition but responds similarly to coherent passages and lists of disconnected sentences. Longer-range discourse, narrative coherence, event schemas, and some situation-model operations recruit systems outside it, especially default-mode and theory-of-mind networks. Some of these systems integrate information over minutes rather than a few words.

For CCF, discourse and situation models should not be stored inside a lexical construction processor. Language can cue and constrain them, while recurrent coordination among conceptual, perceptual, action, memory, and control networks builds the larger model.

The Casto et al. paper further suggests that a situation model may arise from dynamic coordination among several domain-specific systems, perhaps with hippocampal participation, rather than from one universal situation-model module. CCF should test both coordinated specialization and more centralized alternatives instead of presupposing either.

12.6 Revised status of NCL claims

NCL claim	Update from recent evidence	CCF status
Complex language is distributed over connected areas.	Strongly supported; the core network is anatomically and functionally interconnected.	Retain.
Local areas perform more limited functions.	Supported, but functional areas and intermingled populations must be identified functionally, not by coarse anatomy.	Retain and refine.
Recognition and production interact.	Core high-level language regions participate in both; perceptual and motor interfaces remain partly specialized.	Retain with subsystem boundaries.
Written, spoken, and conceptual routes are distinguishable.	Strongly supported by modality-specific fusiform/STG responses and higher amodal responses.	Retain.
Syntax has a dedicated center.	Not an NCL requirement and not supported by the reviewed modern evidence; syntax and lexical semantics are broadly integrated.	Reject fixed syntax-center implementations.
“Broca’s area” produces speech and “Wernicke’s	Too coarse. Articulatory planning, speech	Replace named centers with functional

NCL claim	Update from recent evidence	CCF status
area" understands it.	perception, high-level language, execution, and broader cognition are dissociable.	networks.
Concepts are distributed but may have local cardinal access nodes.	Sparse invariant concept cells and sharply selective linguistic neurons make this plausible, but do not prove neocortical concept cardinals or uniqueness.	Implement as a testable abstraction with optional redundancy.
A cardinal is the unique apex of every web.	Not established; network redundancy and distributed high-level codes caution against mandatory uniqueness.	Treat as a hypothesis, not an invariant.
Conceptual content and lexical form are distinct.	Strongly reinforced by dissociations between the language network and nonlinguistic cognition.	Make architecturally explicit.
All language-relevant cognition is one undifferentiated web.	Recent work finds reproducible specialized networks that interact closely.	Use one connected system with typed functional subnetworks, not one homogeneous processor.
Bidirectional context affects language processing.	Retained; the mechanisms and routing remain open.	Preserve recurrence and test routing versus broadcasting.

13. Cortical columns as a substrate for NCL

13.1 Scope and status of the column materials

This section draws on two additional study artifacts:

- `Comprehensive Documentation_ Cortical Column Circuits and Neural Connectivity.pdf`;
- `Cortical_Column_Circuit_Dynamic.html`.

The PDF assembles a useful multiscale account of cortical layers, pyramidal projection classes, interneuron motifs, predictive coding, plasticity, and intercolumnar communication. However, it does not provide itemized citations tying each numerical value or circuit claim to a primary source. Its exact cell counts, connection probabilities, pathway assignments, and predictive-coding interpretations must therefore be treated as **modeling notes requiring source verification**, not as a settled anatomical specification.

The HTML artifact is explicitly a **conceptual population-level simulation**, not a spiking, conductance-based, or quantitatively calibrated neural model. It uses normalized activation, hand-selected weights and time constants, bounded leaky updates, abstract behavioral-state inputs, and a simplified reward-dependent plasticity scalar. Its scientific value is explanatory: it makes causal ordering, competing pathways, dendritic versus somatic control, and intercolumn interaction visible. It should not be used as evidence that the illustrated equations or timings are biologically measured.

These materials are nevertheless valuable for identifying which causal distinctions a CCF column abstraction may need to preserve.

13.2 Resolve the word "column" before implementation

The sources and NCL use several related but non-identical units:

Term	Intended scale or role
Neuron	Individual excitable cell with dendritic, somatic, axonal, and synaptic dynamics.
Minicolumn/microcolumn	Narrow vertical population spanning cortical layers; often estimated at tens to roughly one hundred neurons, but its exact anatomical and functional status is debated.
Macrocolumn/cortical column	Wider local organization containing many minicolumns and often sharing a broad response preference.
Functional column in NCL	Learned, variable-sized population treated as a relational-network node; it may correspond to one or more anatomical minicolumns rather than a fixed histological object.
CCF simulated column	A future computational abstraction whose relation to these biological scales must be declared and validated.

The organizational direction is:

```
neurons → minicolumns → larger functional columns → areas and distributed networks
```

Columns do not combine to form minicolumns; minicolumns are the smaller candidate building blocks. Larger functional coalitions can, however, recruit many columns across local and distant regions.

For the V7 discussion, **cortical column** should provisionally mean the smallest addressable CCF processing population, inspired primarily by Lamb's functional column. This is an abstraction level, not a claim that one CCF object equals one universally defined anatomical minicolumn or 300–500 µm macrocolumn.

13.3 The biological circuit described by the materials

The PDF presents a canonical six-layer organization:

- **Layer 1:** sparse somata and dense distal dendrites/axons; a major site for contextual and descending input onto apical tufts.
- **Layers 2/3:** recurrent and horizontal intracortical integration, communication with nearby columns, and feedforward projection to other cortical areas.
- **Layer 4:** principal thalamorecipient input layer in granular cortex, with strong local and feedforward inhibitory control.
- **Layer 5:** deep pyramidal output to cortical, striatal, brainstem, and other subcortical targets, with strong recurrent participation.
- **Layer 6:** corticothalamic and local feedback that modulates thalamic relay and cortical input processing.

Its simplified canonical flow is:

```
thalamus → L4 → L2/3 → L5 → cortical/subcortical output
                    ↕
           lateral and recurrent context

higher cortex → L1 apical compartments and deep layers
L6 → thalamic/input modulation
```

The document emphasizes that this is not the only route. Thalamic input can reach other layers, feedforward and feedback projections vary by cortical area, recurrent loops are extensive, and the morphology of nominally similar populations differs across regions.

Three broad inhibitory motifs are used throughout the PDF and artifact:

- **PV-like fast perisomatic inhibition:** controls gain, spike timing, feedforward/feedback inhibition, and narrow temporal windows.
- **SOM-like dendritic inhibition:** gates distal/apical contextual input and can coordinate inhibition across layers.
- **VIP-like disinhibition:** inhibits SOM-like populations, thereby opening selected dendritic compartments under attention, state, novelty, or learning conditions.

This produces a useful functional decomposition:

```
bottom-up evidence enters a local input compartment
+ recurrent and lateral evidence enters an integration population
+ top-down context biases a distinct dendritic compartment
− fast somatic inhibition controls timing and gain
− slower dendritic inhibition controls contextual influence
+ disinhibition selectively opens contextual/plasticity gates
→ deep-layer output and feedback
```

This is richer than a single scalar threshold node. It shows why a local population can be generic while still changing its effective computation according to input route, timing, inhibition, and behavioral state.

13.4 What the dynamic artifact demonstrates

The artifact contains two lower-order columns, a neighboring competitor, a higher-order column, thalamic input, subcortical output, and abstract attention/arousal/novelty/reward populations. Separate nodes represent L4 input, L2/3 integration, L5 output, L6 feedback, L2/3 and L5 apical tufts, and PV/SOM/VIP-like control populations.

It exposes nine pathway views:

1. all circuits together;
2. thalamocortical input;
3. intracolumnar flow;
4. ascending prediction error;
5. top-down prediction;
6. lateral competition;
7. PV-like fast gain control;
8. SOM-like dendritic gating;
9. VIP-like disinhibition.

It composes these pathways into ten process demonstrations:

- sensory integration;
- predictive-coding loop;
- winner-takes-most competition;
- perisomatic gain control;
- combined behavioral-state modulation;
- attention;
- arousal;
- novelty/surprise;
- reward-gated plasticity;
- sustained pyramidal activity.

Several lessons are directly relevant to CCF:

- **A process is not inside one node.** Sensory integration is a temporally ordered interaction among layers, inhibition, neighboring columns, and higher context.
- **Communication is route-sensitive.** Bottom-up, horizontal, top-down, modulatory, inhibitory, and output influences are not interchangeable weighted messages.
- **Context is gated before output.** Descending activity reaches explicit apical compartments and may bias rather than directly determine firing.
- **Competition is iterative.** A small initial advantage is amplified through several reciprocal inhibitory rounds, producing "winner takes most," not an instantaneous symbolic winner.
- **Persistence is bounded.** L2/3–L5 recurrence sustains activity while fast PV-like and slower SOM-like feedback prevent runaway excitation.
- **Plasticity is state-dependent.** In the reward scenario, contextual coincidence changes a connection only while a disinhibitory gate is open.
- **Multiple columns cooperate and compete simultaneously.** A local representation is meaningful because of vertical, lateral, hierarchical, thalamic, and output connectivity.

The artifact update is approximately a bounded leaky-rate model rather than a neural simulation: weighted excitatory, inhibitory, and modulatory effects produce a target activation, and each population relaxes toward that target with a class-dependent time constant. The display adds pulse schedules and moving particles to expose causal order. This distinction must remain explicit when the artifact informs CCF design.

13.5 How column circuitry grounds NCL premises

The circuit account supplies possible substrate-level mechanisms for many NCL operations:

NCL premise or operation	Candidate columnar realization
Connectivity determines function.	A generic local circuit acquires its role from thalamic, cortical, lateral, hierarchical, and output connections.
Incoming activation is integrated and thresholded.	Separate dendritic and somatic integration under balanced excitation and inhibition.
A node broadcasts activation.	Pyramidal populations project horizontally, to other cortical areas, and through deep-layer outputs.
Activation can persist or WAIT.	Recurrent L2/3–L5 excitation maintains a bounded state after the initiating input.
Alternatives compete.	Horizontal excitation recruits somatic and dendritic inhibition in neighboring populations.
Context affects recognition.	Descending input reaches apical compartments and changes the response to bottom-up drive.
Attention selects a web or route.	Disinhibitory control changes which contextual inputs can influence a population.
Functional webs reverberate.	Reciprocal intercolumnar and interareal loops stabilize mutually consistent populations.
Learning strengthens successful conjunctions.	Timing-dependent and Hebbian plasticity is gated by local state, inhibition, neuromodulation, and behavioral outcome.
A cardinal ignites a distributed web.	A highly connected convergence population could broadcast through recurrent routes, while its supporting content remains distributed.

This makes NCL's column less like a Boolean symbol and more like a **stateful, context-sensitive convergence-and-routing population**. AND, OR, WAIT, competition, and precedence would be emergent circuit configurations over columns, not hard-coded instruction opcodes inside them.

It also sharpens the cardinal-node hypothesis. A cardinal's apparent specificity could arise from convergent connectivity, recurrent support, and competitive stabilization. Its activation might help ignite a web through deep and horizontal outputs. But the circuit materials do not establish that every web has one unique apex. Redundant or context-dependent convergence populations remain compatible with both the biology and recent language evidence.

13.6 Relations and constructions remain network-level organizations

A relation such as IN, ACROSS, or TOWARD is not one dedicated wire carrying a symbolic tag. It is a learned condition on how activity is transformed, routed, stabilized, sequenced, or inhibited across populations. A relation cardinal may provide an address for invoking that organization while its realization remains in distributed connectivity.

Likewise, image schemas and grammatical constructions should not be preloaded inside special processors. Sensorimotor activity, recurrent pathways, ordered conjunctions, competition, persistence, and learned convergence must jointly realize them. This preserves NCL's claim that relations connect relations all the way down rather than being interpreted by a separate symbolic mechanism.

13.7 Causal distinctions to preserve before simplification

The next CCF discussion can simplify aggressively, but it should first decide which distinctions are scientifically indispensable. The materials suggest evaluating at least:

1. bottom-up evidence versus top-down/contextual influence;
2. local recurrent state versus current external input;
3. dendritic/context gating versus somatic/output gain;
4. excitatory propagation versus inhibitory competition;
5. local, lateral, hierarchical, and action/output routes;
6. fast response dynamics versus slower persistence and plasticity;
7. ordinary activation versus learning eligibility;
8. autonomous cognitive state versus attention, reward, or other control provenance;

9. single-column state versus coalition-level Functional Web activity;
10. local convergence points versus distributed and redundant representations.

This does **not** require CCF to expose six literal layers, thousands of neurons, oscillations, every interneuron subtype, or the PDF's numerical connection matrix. PV, SOM, and VIP names should not become semantic schema primitives or training labels. If retained, they belong to a substrate model; a simpler implementation may instead preserve their functional consequences as fast gain control, dendritic contextual gating, and disinhibition/plasticity gating.

The theme to carry into the implementation discussion is therefore:

> A CCF cortical column is intended to be the smallest addressable processing population, but represented knowledge is not inside it. Its cognitive role emerges from compartment-sensitive integration, recurrent state, inhibition, plasticity, and connectivity within distributed NCL Functional Webs.

---

14. Implications for CCF Version 7

14.1 The CCF column should be smaller than the CCF 6.0 CM

The current CCF Columnar Module can contain an authored state space, semantic roles, transitions, completion conditions, and ports. Lamb's columnar node is more primitive:

```
activation state
threshold/response function
persistence
incoming/outgoing weighted connections
local competition
plasticity/recruitment status
```

Schemas, constructions, words, and concepts should be **functional webs made from such units**, not different large programs stored inside one column.

14.2 A cardinal is an ordinary recruited column in a special network position

A cardinal need not be a different processor type. It is a column—or a small redundant population of columns—that has become a stable convergence point for a web. Its role comes from connectivity and learning. Being the unique highest point is a hypothesis to test, not a required architectural invariant.

This suggests that CCF should model:

- recruitment of cardinals;
- subweb cardinals and whole-web cardinals;
- category-to-instance links such as TOM → CAT;
- asymmetric generic-to-instance activation;
- ignition from any sufficiently informative subweb;
- graceful degradation when one path or cardinal is lesioned.

14.3 Functional webs should overlap

A CCF concept, relation, schema, or construction should not necessarily own all its components. Shared properties, lexical parts, relations, actions, and perceptual patterns can participate in many webs.

A "web" may therefore be a traceable activation/provenance view over one global graph rather than a permanently encapsulated module object.

14.4 Keep levels of description explicit

A suitable CCF stack could be:

Level	Description
Cognitive	CAT, TOM, CONTAINER, construction, lexical meaning
Abstract relational	AND/OR, category, sequence, functional web, cardinal
Narrow relational	Directed excitatory/inhibitory nodes, waits, gates, competitors
Columnar	Simulated columns, local state, thresholds, persistence, plasticity
Numerical	Arrays, sparse graphs, equations, ticks, seeds

Compilation must translate between levels without making cognitive labels control numerical behavior.

14.5 NCL must determine the architecture directly

CCF should test NCL's own proposed mechanisms without adding a separate processor for structural state or binding. Cardinal convergence, hierarchically organized Functional Webs, lexical-form/meaning separation, bidirectional recognition and production, morphology, grammar, proximity, competition, entrenchment, and recruitment must be represented through one connected network of ordinary Columns.

The architecture may distinguish sensory, linguistic, conceptual, motor, and control populations when evidence supports those functional differences. Those populations remain connected parts of one system, and learned functions may overlap them. No population should receive semantic authority merely from its configured name or position.

14.6 Suggested minimum V7 column abstraction

The next discussion can evaluate a unit with at least:

```
Column
  stable internal identity for simulation/provenance
  activation ∈ [0,1]
  graded response/threshold function
  short-term recurrent persistence
  excitatory outgoing connections
  inhibitory local/competitive connections
  incoming integration
  learning/recruitment state
  optional local neighborhood position
```

It should not contain:

- a symbolic concept definition;
- an entire image schema;
- schema-specific execution code;
- a list of inherited properties copied from a parent;
- world coordinates as semantic truth;
- a hard-coded biological response class.

---

15. Scientific cautions

The presentations provide a coherent theory, and recent findings strengthen several premises, but some claims must remain hypotheses in CCF documentation.

1. **Universal columnar node:** columnar organization is influential but its definition, universality, and computational role remain debated. A simulated CCF column should be called an abstraction, not a literal verified cortical unit.
2. **Uniform response within a minicolumn:** recent single-neuron work demonstrates sharply selective cells and locally correlated populations, but does not show that every neuron in a minicolumn has the same response. The original uniformity claim remains too strong.
3. **Conceptual categories mapped to nested columns:** this is an extrapolation from sensory maps, not an established anatomical mapping.
4. **Cardinal nodes:** sparse invariant concept cells and selective linguistic neurons provide partial support for local access points embedded in distributed codes. They do not directly identify a unique neocortical CAT or GRANDMOTHER cardinal column, establish a hierarchy apex, or show that one node ignites the complete web.
5. **Classical language areas:** coarse Broca/Wernicke centers should be replaced by individually variable functional networks. Modern evidence distinguishes speech perception, orthographic perception, modality-independent high-level language, articulatory planning, motor execution, and extra-linguistic cognition, while preserving extensive interaction among them.
6. **Exact anatomical quantities:** estimates of neurons, synapses, minicolumns, and area capacities vary across methods and should not become fixed simulator constants without current review.
7. **Long-distance excitation and local inhibition:** this is a useful coarse abstraction, not an exceptionless wiring law.
8. **Proximity principle:** wiring economy and topography are real constraints, but semantic similarity does not imply a simple two-dimensional cortical neighborhood in every case.
9. **Hebbian sufficiency:** Hebbian recruitment alone does not yet explain credit assignment, inhibition learning, consolidation, pruning, or acquisition of arbitrary complex constructions.
10. **Language effects on thought:** the proposed network mechanism is plausible, but individual experimental claims require independent replication and careful interpretation.
11. **Core-language-network boundaries:** the Fedorenko et al. synthesis provides strong evidence for functional specialization, but the exact borders may be graded, and other researchers dispute how narrowly language should be localized. CCF should model the reported dissociations without assuming impermeable modules.
12. **Recent preprints:** the single-neuron and exportation papers are not equivalent to settled consensus. Their claims should be represented as provisional evidence and explicit hypotheses, respectively.
13. **Column terminology:** minicolumn, microcolumn, macrocolumn, cortical column, and functional column do not identify one universally accepted anatomical/computational unit. CCF must declare which abstraction it implements rather than combine their measurements.
14. **Canonical microcircuit:** six-layer feedforward diagrams are useful summaries, but real connectivity varies by species, cortical area, layer, cell subtype, development, and behavioral state. The canonical pathway is not a universal fixed circuit.
15. **Interneuron categories:** PV, SOM, and VIP motifs capture important inhibitory functions but do not exhaust cortical interneuron diversity, and their effects are more context-dependent than three fixed gates imply.
16. **Predictive-coding anatomy:** assigning prediction, error, precision, and attention to particular layers and cell classes is a family of

hypotheses, not a settled decoding of cortical circuitry.

17. **Numerical parameters in the column PDF:** exact counts, percentages, distances, delays, oscillation bands, and connection probabilities need traceable primary sources and area-specific provenance before use as simulator defaults.
18. **Dynamic HTML artifact:** its normalized activation, scheduled pulses, class-dependent time constants, and reward plasticity are explanatory design choices. It demonstrates qualitative causal motifs but provides no parameter fit or empirical validation.

CCF should therefore classify each adopted idea as an implementation abstraction, theoretical commitment, empirical constraint, provisional result, illustrative mechanism, or open hypothesis.

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16. Presentation-by-presentation guide

Deck	Main contribution
`Ling411-01`	Defines linguistic neuroscience, the individual linguistic system, and operational/developmental/neurological plausibility; introduces distributed cortical language organization.
`Ling411-02-Neurons`	Neurons, synapses, excitation/inhibition, integration, cortical layers, white-matter connections, primary/association hierarchy, and connectivity as the basis of function.
`Ling411-03-History`	Aphasiology from early localization through Broca, Wernicke, Lichtheim, Geschwind, and modern criticism; basic versus complex functions.
`Ling411-04-Symptoms`	Aphasic behavior, agrammatism, anomia, paraphasia, and the decomposition of speaking and comprehension into interacting processes.
`ling411-05-Types`	Wernicke-type syndromes and the limitations of treating aphasia classes as sharply discrete.
`ling411-06-Classification_Laterality`	Competing aphasia classifications, anterior/posterior distinctions, hemispheric laterality, category-specific semantic deficits, and bilateral meaning.
`ling411-07-Localization`	Lesion variability, cautious localization inference, reading/writing pathways, and partially independent graphic, phonological, and semantic routes.
`ling411-08-Grammar`	Agrammatism, relational markers, imagery, bidirectional processing, and reasons not to identify grammar with one cortical center.
`ling411-09-Imaging`	Mapping methods and their limitations: lesions, stimulation, TMS, EEG/ERP, PET, and fMRI.
`ling411-10-MEG`	MEG/MSI, temporal dynamics, source localization, methodological limits, and transition into network/columnar anatomy.
`ling411-11-Columns`	Mountcastle-based column hierarchy, local operations, specificity, adjacency, proximity, competition, and the cortex as a network of columns.
`ling411-12-LingNeuroscience`	Extrapolates perceptual column principles to language; develops integration, broadcasting, persistence, sequence recognition, and symbol-free linguistic processing.
`ling411-13-FunctionalWebs`	Distributed functional webs, distinct subwebs, hierarchy, cardinal nodes, ignition, lexical form/meaning organization, and local-plus-distributed representation.
`ling411-14-RH`	Further arguments for cardinal nodes, reverberation, the arbitrary sign relation, and right-hemisphere contributions.
`ling411-15-Questions`	Clarifies cardinal-node questions, functional webs versus the global network, category deficits, overlapping webs, and shared components.
`ling411-16-LinguisticEvidence`	Derives relational networks from linguistic evidence; abstract/narrow notation, constructions, WAIT, competition, morphology, lexicalization, polysemy, and recurring semantic components.
`ling411-17-Perception`	Macroperception as multimodal, top-down, motor-involving, and language-sensitive; reciprocal activation and inference.
`ling411-18-CognitiveMaps`	Conceptual and phonological map hypothesis, prototypes, fuzzy categories, hierarchical columnar subdivision, and category learning.
`ling411-19-Learning`	Hebbian success rule, latent abundance, recruitment, threshold adjustment, supercolumn subdivision, proximity, inherited constraints, and entrenchment.
`ling411-21-FunctNeuroanat`	Critiques category and either/or thinking in functional neuroanatomy; interprets computation as integration and broadcasting.
`ling411-22-FunctNeuroanat-II`	Bilateral speech perception, perception–production interfaces, area Spt, tractography, and alternative direct/intermediate pathway models.

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17. Compact synthesis

Neurocognitive Linguistics proposes that an individual's linguistic-cognitive system is one dynamic physical network. Recent evidence refines this into distinguishable but tightly interacting functional subnetworks: modality-specific perceptual interfaces, an amodal high-level language network, production-planning and motor interfaces, and extra-linguistic systems for grounded knowledge, reasoning, memory, and situation models. This refinement preserves NCL's relational and distributed premises while rejecting coarse anatomical centers and a homogeneous whole-brain processor.

Cortical columns remain candidate units for local integration, persistence, broadcasting, inhibition, and plasticity. Their functions should come from connectivity rather than internal symbols. Learned words, concepts, constructions, and percepts can be represented as overlapping functional webs. Compact cardinal convergence populations may provide stable addressability, routing, and reactivation while leaving content distributed, but their uniqueness, anatomical scale, and necessity must be tested. Bidirectional activation supports recognition, production, imagery, inference, and completion. Weighted connections and graded thresholds explain entrenchment, prototypes, polysemy, and fuzzy categories. Learning recruits abundant latent structure and progressively differentiates broad populations into finer functional units.

For CCF Version 7, the most consequential requirement is:

> **Do not model a cortical column as a container holding a concept or schema. Model it as a small stateful convergence-and-broadcast unit. Concepts, schemas, constructions, lexical structures, and situation models should arise through learned functional webs spanning appropriately specialized systems. Treat cardinal columns or cardinal populations as testable local access mechanisms for distributed content, not as pre-established unique concept stores.**